

RIKESH MANDAL

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PROFESSIONAL PROFILE

Software Engineer with an MSc in Artificial Intelligence (Distinction, 82%) and a First Class BSc in Computer Networking and IT Security (81.94%). Two years of professional experience designing and maintaining Java and Spring Boot REST APIs for a regulated fintech platform processing real-time card authorisations. Skilled in backend development, API design, Swagger/OpenAPI documentation, unit and integration testing, with Git-based version control and CI/CD workflows. Additional applied experience in Python, machine learning (PyTorch, TensorFlow, Scikit-learn), and computer vision from postgraduate research. Strong interest in QA, DevOps, with foundational exposure to Docker, AWS, and automated testing principles.

KEY SKILLS

Programming Languages: Java, Python, JavaScript, MATLAB

Backend & Web Development: Spring Boot, Flask, Django, REST APIs, Java Spark, Jinja2, HTML, CSS

APIs & Messaging: REST API Design, Swagger/OpenAPI, Postman, JSON, NATS, Event-Driven Architecture, WebSockets

Databases: MySQL, PostgreSQL, SQLite, MongoDB, SQLAlchemy, Database Indexing, Query Optimisation

Frontend: Vue.js, HTML, CSS, JavaScript

Testing & QA: Unit Testing, Integration Testing, Pytest, Test Case Design, Bug Reproduction & Documentation, Postman

CI/CD & DevOps: Git, GitHub, CI/CD Pipelines, Docker (foundational), Linux, AWS (Academy Cloud Foundations)

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Open3D

Databases: MySQL, PostgreSQL, SQLite, MongoDB

PROFESSIONAL EXPERIENCE

Assistant Java Developer, FocusOne Payment Solutions (Fintech)

2022 – 2024

Backend systems powering real-time credit and debit card authorisation for a regulated fintech platform

- Designed and maintained high-performance REST APIs using Java and Spring Boot for a platform processing real-time card authorisations across thousands of daily transactions.
- Refactored legacy applications through root-cause debugging and targeted code improvements, reducing production incidents and improving system reliability.
- Evaluated NATS messaging for event-driven architecture adoption and presented findings that shaped the team's infrastructure roadmap for low-latency, fault-tolerant systems.
- Built and maintained Swagger/OpenAPI documentation and contributed to API endpoint testing using Postman, cutting onboarding time and improving cross-team collaboration on API contracts.
- Delivered new backend features aligned with evolving product requirements in a fast-paced Agile environment, participating in code reviews and sprint planning.
- Authored system specifications and API design documentation supporting long-term maintainability, reliability, and regulatory compliance.

KEY PROJECTS

Microblog — Full-Stack Flask Web Application | [GitHub](#) | [Live Demo](#)

Built and progressively extended a full-stack Flask web application based on Miguel Grinberg's Flask Mega-Tutorial, implementing authentication, user management, database-driven features, search, translation, notifications, background processing, testing, containerisation and cloud deployment.

- Implemented user authentication, registration, profiles, following/followers, notifications, password reset,

forms and server-side validation using Flask, Flask-Login and related extensions.

- Designed relational data models using SQLAlchemy ORM, working with SQLite and MySQL, database relationships, foreign keys, migrations with Flask-Migrate/Alembic, database indexing and query optimisation.
- Implemented pagination for efficiently handling larger collections of posts and users.
- Developed REST-style endpoints and JSON-based asynchronous interactions, using JavaScript for dynamic UI functionality, pop-ups and client-side requests.
- Implemented and explored polling and WebSockets for real-time application updates and communication.
- Integrated Elasticsearch for full-text search and implemented database commit hooks to synchronise searchable records with the search index.
- Integrated LibreTranslate for multilingual post translation and implemented internationalisation using Flask-Babel.
- Implemented background task processing concepts using Redis and RQ for handling work outside the request/response cycle.
- Containerised the application and supporting services using Docker and Docker Compose, including container networking, environment variables and persistent MySQL and Elasticsearch volumes.
- Deployed the Flask application and MySQL database to Railway, configuring production environment variables and database connectivity.
- Elasticsearch and LibreTranslate are currently self-hosted for development/testing and are not included in the public Railway deployment.

Distributed Messaging System with NATS.io (Java Microservices)

- Led the design and implementation of a distributed microservices communication system using NATS.io, taking responsibility for architecture, message design, and system integration strategy.
- Built pub-sub based request-reply workflows in Java, handling concurrent request processing via multi-threaded responders to improve responsiveness under load.
- Implemented queue groups to balance requests across multiple service instances, improving reliability and preventing bottlenecks.
- Defined JSON-based message contracts enabling independently built services to communicate without tight coupling, and tested the system under concurrent load to validate stability.

Monocular Image-to-3D Reconstruction (MSc Dissertation)

[Github](#)

- Built an end-to-end deep learning pipeline using a ResNet-50 encoder-decoder for monocular depth estimation, converting depth maps to 3D point clouds via Open3D with Poisson surface reconstruction and mesh refinement.
- Tech stack: Python, PyTorch, OpenCV, Open3D, NumPy.

Sign Language-to-Speech Glove

[Github](#)

- Built a hardware-software IoT system using an Arduino glove with flex sensors and a gyroscope to capture hand gestures, and trained a KNN classifier on a self-collected dataset to translate ASL gestures into real-time speech.
- Tech stack: Python, Arduino, Scikit-learn, IoT sensors.

Animal Image Classification System

[Github](#)

- Designed a custom CNN from scratch (no pre-trained weights) to classify 9 animal classes, achieving 79.51% accuracy using progressive convolutional depth, 3x3 filters, batch normalisation, and dropout.
- Tech stack: Python, PyTorch, OpenCV, NumPy, Scikit-learn.

EDUCATION

MSc Artificial Intelligence — Distinction (82%)

2024 – 2025

University of East London

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Data Analytics Specialisation — Coursera (2023)
- API Designer & API Security Architect — API Academy
- AWS Academy Cloud Foundations — Amazon Web Services
- Deep Learning Onramp & Machine Learning Onramp — MathWorks
- Data Science Foundations — LinkedIn Learning